

The mural problem

OR 6205 Deterministic Operations Research · Northeastern University · Spring 2026 · in-class formulation exercise

Four artists are assigned to three murals over an eight-week project. Each artist is better at some murals than others, each mural needs a minimum number of hours before it counts as finished, paint costs differ by mural, and the budget is fixed. Students decide how many hours each artist spends on each mural. The formulation is the exercise; students build it on the board, in groups, before anything is solved.

Sets and Indices

$i \in \{1,2,3,4\}$ = Artists (Alex, Bella, Carlos, Diana)

$j \in \{1,2,3\}$ = Murals (Historical, Nature, Abstract)

Parameters

	Mural 1	Mural 2	Mural 3
Artist 1 (Alex)	9	5	6
Artist 2 (Bella)	6	9	4
Artist 3 (Carlos)	4	6	9
Artist 4 (Diana)	7	7	7

Artist weekly availability (hours):

$a_1=12, a_2=8, a_3=15, a_4=10$

Quality contribution q_{ij} (skill match of artist i on mural j):

Minimum hours required per mural:

$m_1=120, m_2=100, m_3=80$

Paint cost per hour worked on each mural:

$c_1=8, c_2=12, c_3=10$

Total budget: $B=4000$

Project duration: $T=8$ weeks

Decision Variables

x_{ij} = Total hours artist i works on mural j over the 8-week project

where $x_{ij} \geq 0$ for all i,j

Objective Function

Maximize total quality-weighted hours:

$$\max Z = \sum_{i=1}^4 \sum_{j=1}^3 q_{ij} \cdot x_{ij}$$

Expanded:

$$\max Z = 9x_{11} + 5x_{12} + 6x_{13} + 6x_{21} + 9x_{22} + 4x_{23} + 4x_{31} + 6x_{32} + 9x_{33} + 7x_{41} + 7x_{42} + 7x_{43}$$

Constraints

1. Artist availability constraints (total hours over 8 weeks):

$$\sum_{j=1}^3 x_{ij} \leq a_i \times T \text{ for each artist } i$$

$$\text{Alex: } x_{11} + x_{12} + x_{13} \leq 96$$

$$\text{Bella: } x_{21} + x_{22} + x_{23} \leq 64$$

$$\text{Carlos: } x_{31} + x_{32} + x_{33} \leq 120$$

$$\text{Diana: } x_{41} + x_{42} + x_{43} \leq 80$$

2. Mural completion constraints (minimum hours required):

$$\sum_{i=1}^4 x_{ij} \geq m_j \text{ for each mural } j$$

$$\text{Historical: } x_{11} + x_{21} + x_{31} + x_{41} \geq 120$$

$$\text{Nature: } x_{12} + x_{22} + x_{32} + x_{42} \geq 100$$

$$\text{Abstract: } x_{13} + x_{23} + x_{33} + x_{43} \geq 80$$

3. Budget constraint:

$$\sum_{i=1}^4 \sum_{j=1}^3 c_j \cdot x_{ij} \leq B$$

Expanded:

$$8(x_{11} + x_{21} + x_{31} + x_{41}) + 12(x_{12} + x_{22} + x_{32} + x_{42}) + 10(x_{13} + x_{23} + x_{33} + x_{43}) \leq 4000$$

4. Non-negativity:

$$x_{ij} \geq 0 \text{ for all } i, j$$

Summary of the LP

Variables: 12 decision variables (x_{ij} for 4 artists $\times$ 3 murals)

Objective: Maximize $9x_{11} + 5x_{12} + 6x_{13} + 6x_{21} + 9x_{22} + 4x_{23} + 4x_{31} + 6x_{32} + 9x_{33} + 7x_{41} + 7x_{42} + 7x_{43}$

Subject to:

4 artist availability constraints ($\leq$)

3 mural completion constraints ($\geq$)

1 budget constraint ($\leq$)

12 non-negativity constraints

Total: 8 functional constraints + 12 non-negativity constraints